

# SYDNEY TECHNICAL HIGH SCHOOL

( Established 1911 )



## TRIAL HIGHER SCHOOL CERTIFICATE

### 2010

# Mathematics Extension 1

#### General Instructions

- Reading time - 5 minutes
- Working time - 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Diagrams are not drawn to scale

Total marks - 84

- Attempt Questions 1 – 7
- All questions are of equal value

Name : \_\_\_\_\_

Teacher : \_\_\_\_\_

| Question<br>1 | Question<br>2 | Question<br>3 | Question<br>4 | Question<br>5 | Question<br>6 | Question<br>7 | Total |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|-------|
|               |               |               |               |               |               |               |       |

**Question 1** (12 marks)

- (a) Simplify  $\frac{\sin(x+y)+\sin(x-y)}{\cos(x+y)-\cos(x-y)}$  2
- (b) Differentiate  $\cos^{-1}\left(\frac{2x}{3}\right)$  2
- (c) Solve  $\frac{2x}{x-1} < 3$  3
- (d) Find the acute angle between the lines 2  
 $x - 2y + 3 = 0$  and  $3x + y + 6 = 0$   
( Answer to the nearest degree )
- (e) Evaluate  $\int_0^{\sqrt{3}} \frac{4}{x^2+9} dx$  3

**Question 2** (12 marks) Start a new page.

- (a) Use the substitution  $t = \tan \frac{A}{2}$  to simplify  $1 + \tan A \tan \frac{A}{2}$ . 2
- (b) Find the value of  $a$  if the polynomial  $p(x) = x^3 - 2x^2 - ax + 6$  is divisible by  $(x + 2)$ . 2
- (c)  $A(x^2, 12)$  and  $B(x, 6)$  are two fixed points for some real value of  $x$ . 2  
The point  $P(1, 10)$  divides the interval AB internally in the ratio 1:2.  
Find the possible values of  $x$ .
- (d) (i) State the domain and range of  $y = 4 \sin^{-1} \frac{x}{2}$ . 2  
(ii) Sketch  $y = 4 \sin^{-1} \frac{x}{2}$ . 1  
(iii) Find the area bounded by  $y = 4 \sin^{-1} \frac{x}{2}$ , the x axis 3  
and the line  $x = 1$ .

**Question 3** (12 marks) Start a new page.

- (a) Solve  $\sin 4\theta = \cos 2\theta$  for  $0 \leq \theta \leq \pi$ . 3
- (b)  $P(4p, 2p^2)$  is a point on the parabola  $x^2 = 8y$ .
- (i) Find the coordinates of S, the focus of the parabola  $x^2 = 8y$ . 1
- (ii) Find the equation of the tangent at P. 2
- (iii) The tangent at P meets the y axis at the point M. 1
- Find the coordinates of M.
- (iv) The perpendicular from S to the tangent PM meets the 3
- tangent at N. Find the coordinates of N.
- (v) Find the equation of the locus of the midpoint of 2
- the interval MN as the position of P varies.

**Question 4** (12 marks) Start a new page.

- (a) The volume of a cube is expanding at the constant rate of  $5 \text{ mm}^3/\text{sec}$ . 4
- At what rate is the surface area of the cube increasing when
- the side length of the cube is 60 centimetres ?

**Question 4 continues on page 5**

Question 4 ( continued )

- (b) A particle is moving along the  $x$  axis. 2

Its velocity  $v$  at position  $x$  is given by  $v = 12 - x^2$ .

Find the acceleration of the particle when  $x = 4$ .

- (c) From point A, Sarah observed that the base of a tower is at a 3

bearing of  $080^\circ$  and the top of the tower is at an angle of

elevation of  $9^\circ$ . Sarah then walks to point B, 1000m due South of A,

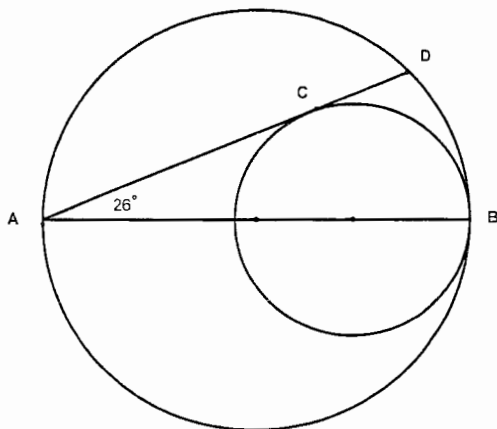
and observes that the base of the tower is at a bearing of  $065^\circ$ .

Find the height of the tower above ground level.

( Give answer in metres correct to 1 decimal place.)

( Points A and B are at ground level. )

- (d)



The diameter AB of the larger circle is 10 centimetres. 3

The smaller circle touches the larger circle at B and the

chord ACD is a tangent to the smaller circle. Angle  $DAB = 26^\circ$ .

Find the radius of the smaller circle, in centimetres correct to 2 decimal places.

**Question 5** (12 marks) Start a new page.

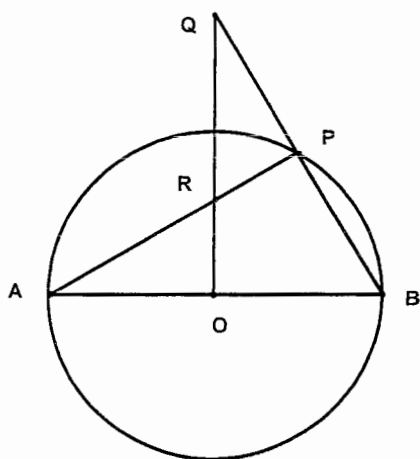
(a) Find  $\int \cos^2 4x \, dx$  2

(b) If  $f(x) = \log_e \sqrt{5 - 2x}$  find the inverse function  $f^{-1}(x)$ . 2

(c) O is the centre of the circle.

BPQ, ORQ, ARP and AOB are straight lines.

Angle QOB =  $90^\circ$ .



(i) Copy the diagram onto your answer sheet.

(ii) Prove that A, O, P and Q are concyclic points. 3

Question 5 continues on page 7

Question 5 ( continued )

- (d) The temperature of a liquid  $t$  minutes after being placed in a freezer is given by the equation  $T = -4 + Ae^{-kt}$ , where  $A$  and  $k$  are constants.

- (i) Initially the liquid is at a temperature of  $40^{\circ}\text{C}$ . 1

Find the value of  $A$ .

- (ii) When the temperature of the liquid is  $26^{\circ}\text{C}$  the rate of change of the temperature of the liquid is  $-0.3^{\circ}\text{C}$  per minute. 2

Show that  $k = 0.01$ .

- (iii) Find the time taken for temperature of the liquid to fall from  $40^{\circ}\text{C}$  to  $6^{\circ}\text{C}$ . (Answer in minutes correct to 1 decimal place) 2

**Question 6** (12 marks) Start a new page.

(a) Find  $\int \frac{dx}{2x\sqrt{1-(\ln x)^2}}$  using the substitution  $u = \ln x$ . 3

(b) A particle is projected from ground level with a velocity of 32 m/s.

The angle of elevation,  $\theta$ , is allowed to vary.

You may assume that, if the origin is taken to be the point of projection,

the path of the particle at time  $t$  seconds is given by the parametric equations

$$x = 32t\cos\theta$$

$$y = 32t\sin\theta - \frac{1}{2}gt^2 \quad \text{where } g \text{ m/s}^2 \text{ is the acceleration due to gravity.}$$

(i) Show that the maximum height reached by the projectile 2

is given by  $\frac{512 \sin^2\theta}{g}$  metres.

(ii) Find an expression for the maximum distance the particle 3

can land from the point of projection.

(iii) The particle is to be projected so as to hit an object 30 metres above 4

ground level and 64 metres horizontally from the point of projection.

Taking  $g = 10 \text{ m/s}^2$ , calculate the possible angles of projection.

(Give answers correct to the nearest degree)



**Question 7** (12 marks) Start a new page.

- (a) Use Mathematical Induction to show that

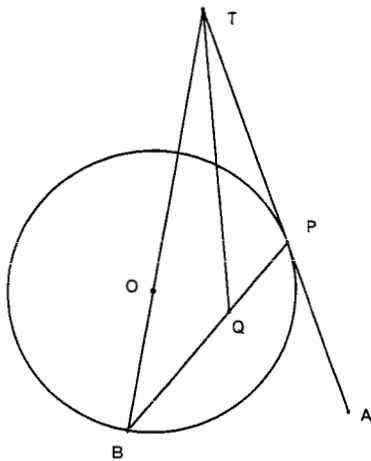
$$5^n \geq 1 + 4n \text{ for all positive integers } n. \quad 3$$

- (b) One root of the equation  $x^3 + px^2 + qx + r = 0$  equals the sum of the other two roots. 3

Prove that  $p^3 = 4pq - 8r$ .

Prove that  $p^3 = 4pq - 8r$ .

- (c)



$P$  is a point on a circle.  $TP$  is a tangent to a circle, centre  $O$ . 3

$TOB$  and  $TPA$  are straight lines.  $QT$  bisects angle  $BTP$ .

Let angle  $PTQ = x$ .

Copy the diagram onto your answer sheet and then

find an expression for angle  $APB$ , giving reasons for each step.

- (d) Find all real  $x$  such that  $|2x - 1| > \sqrt{x(2 - x)}$  3

**End of Paper. ➔**

## STHS EXT 1 TRIAL SOLUTIONS

### Question 1

$$a) \frac{\sin x \cos y + \sin y \cos x + \sin x \cos y - \sin y \cos x}{\cos x \cos y - \sin x \sin y - \cos x \cos y - \sin x \sin y}$$

$$= \frac{2 \sin x \cos y}{-2 \sin x \sin y}$$

$$= -\cot y$$

$$b) \frac{-2}{\sqrt{9-4x^2}}$$

$$c) x \neq 1 \quad 2x = 3x - 3$$

$$x = 3$$



$$\therefore x < 1 \text{ or } x > 3$$

$$d) m_1 = \frac{1}{2} \quad m_2 = -3$$

$$\tan \theta = \left| \frac{-3 - \frac{1}{2}}{1 + (-3)(\frac{1}{2})} \right|$$

$$\therefore \theta = 82^\circ$$

$$e) \left[ \frac{4}{3} \tan^{-1} \frac{x}{3} \right]_0^{\sqrt{3}}$$

$$= \frac{4}{3} \left( \tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right)$$

$$= \frac{4}{3} \cdot \frac{\pi}{6}$$

$$= \frac{2\pi}{9}$$

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Question 2

$$\begin{aligned}
 \text{a)} \quad & 1 + \tan A \tan \frac{A}{2} \\
 &= 1 + \frac{2t}{1-t^2} \cdot t \\
 &= \frac{1-t^2 + 2t^2}{1-t^2} \\
 &= \frac{1+t^2}{1-t^2} \\
 &= \sec A
 \end{aligned}$$

$$\text{b)} \quad p(-2) = 0$$

$$\begin{aligned}
 \therefore -8 - 8 + 2a + 6 &= 0 \\
 a &= 5
 \end{aligned}$$

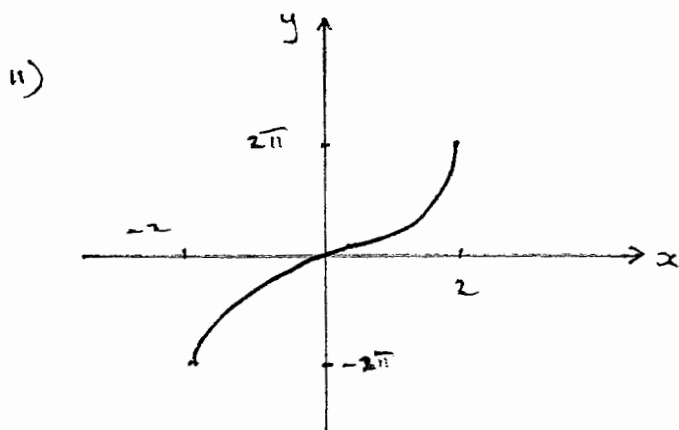
$$\text{c)} \quad \frac{2x^2 + 1x}{3} = 1$$

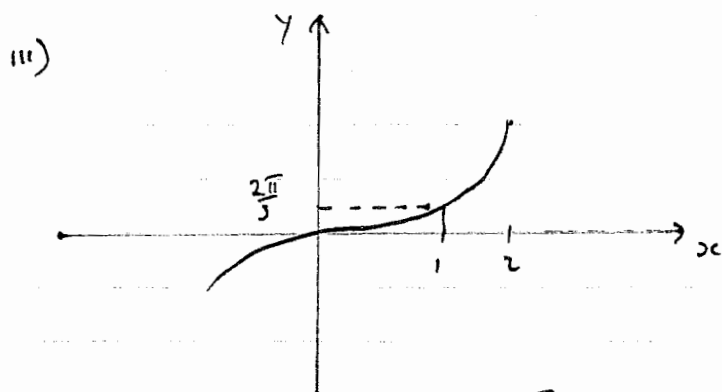
$$2x^2 + x - 3 = 0$$

$$(2x+3)(x-1) = 0$$

$$x = -\frac{3}{2}, +1$$

$$\begin{aligned}
 \text{d)} \quad \text{i)} \quad & D: -2 \leq x \leq 2 \\
 & R: -2\pi \leq y \leq 2\pi
 \end{aligned}$$





$$y = 4 \sin^{-1} \frac{x}{2}$$

$$x = 2 \sin \frac{y}{4}$$

$$\begin{aligned} \text{Area} &= \frac{2\pi}{3} - \int_0^{\frac{2\pi}{3}} 2 \sin \frac{y}{4} dy \\ &= \frac{2\pi}{3} + \left[ 8 \cos \frac{y}{4} \right]_0^{\frac{2\pi}{3}} \\ &= \frac{2\pi}{3} + 8 \cdot \frac{\sqrt{3}}{2} - 8 \\ &= \frac{2\pi}{3} + 4\sqrt{3} - 8 \text{ sq units} \end{aligned}$$

### Question 3

a)  $2 \sin 2\theta \cos 2\theta - \cos 2\theta = 0$   
 $\cos 2\theta (2 \sin 2\theta - 1) = 0$   
 $\cos 2\theta = 0 \quad \sin 2\theta = \frac{1}{2}$

$$2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{12}, \frac{5\pi}{12}$$

b) i)  $S(0, 2)$

ii)  $y = \frac{x^2}{8}$

$$\frac{dy}{dx} = \frac{x}{4} \quad \text{when } x = 4p$$

$$m_T = p$$

$$\therefore y - 2p^2 = p(x - 4p)$$

$$y = px - 2p^2$$

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$$\text{iii) sub } x=0 \quad y = -2p^2$$

$$\therefore M(0, -2p^2)$$

$$\text{iv) } m_{\perp} = -\frac{1}{p}$$

$$\therefore y - 2 = -\frac{1}{p}(x - 0)$$

$$py - 2p = x$$

$$y = 2 - \frac{1}{p}x$$

Solve simultaneously with tangent

$\therefore$

$$2 - \frac{1}{p}x = px - 2p^2$$

$$x\left(p + \frac{1}{p}\right) = 2(p^2 + 1)$$

$$x = \frac{2(p^2 + 1)}{\frac{p^2 + 1}{p}}$$

$$x = 2p$$

$$\therefore y = 2p^2 - 2p$$

$$= 0$$

$$\therefore N(2p, 0)$$

$$\text{v) midpoint } (p, -p^2)$$

$$\therefore x = p, \quad y = -p^2$$

$$\therefore y = -x^2$$

Question 4

a)  $V = x^3$        $A = 6x^2$

$$\frac{dV}{dt} = \frac{dV}{dx} \cdot \frac{dx}{dt}$$

$$\therefore 5 = 3x^2 \cdot \frac{dx}{dt}$$

$$\frac{dx}{dt} = \frac{5}{3x^2}$$

$$\frac{dA}{dt} = \frac{dA}{dx} \cdot \frac{dx}{dt}$$

$$= 12x \times \frac{5}{3x^2}$$

$$= \frac{20}{x}$$

when  $x = 600 \text{ mm}$

$$\frac{dA}{dt} = \frac{1}{30} \text{ mm}^2/\text{sec}$$

b)  $\frac{1}{2}v^2 = \frac{1}{2}(12 - x^2)^2$

$$\therefore a = \frac{d}{dx} \left( \frac{1}{2}v^2 \right)$$

$$= -2x(12 - x^2)$$

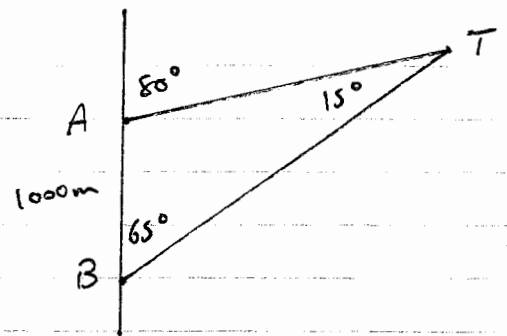
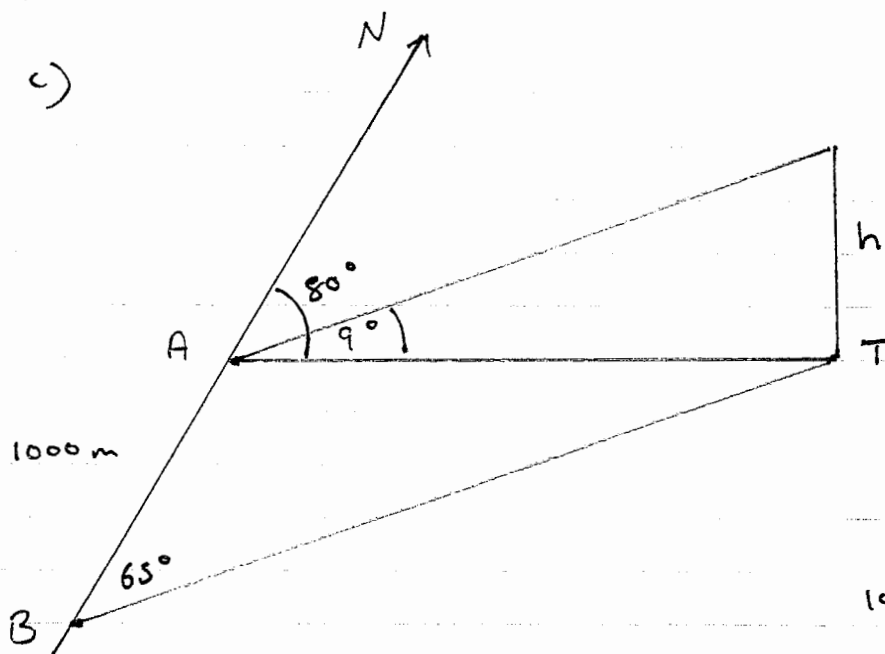
when  $x = 4$

$$a = -8(12 - 16)$$

$$= 32$$

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c)



$$\tan 9^\circ = \frac{h}{AT}$$

$$AT = h \tan 81^\circ$$

$$\therefore \frac{h \tan 81^\circ}{\sin 65^\circ} = \frac{1000}{\sin 15^\circ}$$

$$h = \frac{1000 \sin 65^\circ}{\sin 15^\circ \tan 81^\circ}$$

$$= 554.6 \text{ m}$$

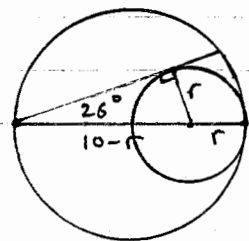
d)

$$\sin 26^\circ = \frac{r}{10-r}$$

$$10 \sin 26^\circ = r (1 + \sin 26^\circ)$$

$$r = \frac{10 \sin 26^\circ}{1 + \sin 26^\circ}$$

$$= 3.05 \text{ cm}$$



Question 5

$$a) \int \cos^2 4x \, dx$$

$$= \frac{1}{2} \int (1 + \cos 8x) \, dx$$

$$= \frac{1}{2} \left( x + \frac{1}{8} \sin 8x \right) + c$$

$$\cos 2x = 2 \cos^2 x - 1$$

$$\cos^2 x = \frac{1}{2} (1 + \cos 2x)$$

$$\cos^2 4x = \frac{1}{2} (1 + \cos 8x)$$

$$b) \quad y = \log_e (5 - 2x)^{\frac{1}{2}}$$

$$\therefore \text{inverse is} \quad x = \log_e (5 - 2y)^{\frac{1}{2}}$$

$$x = \frac{1}{2} \log_e (5 - 2y)$$

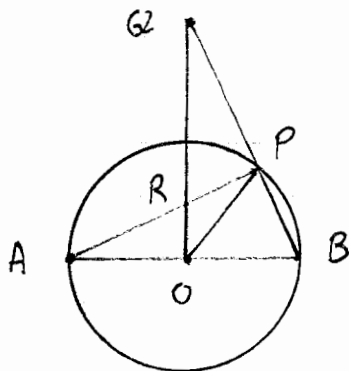
$$2x = \log_e (5 - 2y)$$

$$e^{2x} = 5 - 2y$$

$$y = \frac{1}{2} (5 - e^{2x})$$

$$\therefore f^{-1}(x) = \frac{1}{2} (5 - e^{2x})$$

c) i)



$$ii) \text{ let } \angle APO = x$$

$$\therefore \angle PAO = x \text{ (equal angles in isosceles triangle)}$$

$$\angle APB = 90^\circ \text{ (angle in a semi-circle)}$$

$$\angle PBA = 90^\circ - x \text{ (angle sum of triangle)}$$

$$\angle OQB = x \text{ (angle sum of triangle)}$$

$\therefore A, O, P, Q$  concyclic (OP subtends equal angles at A and Q)



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$$d) \quad i) \quad T = -4 + Ae^{-kt}$$

$$\text{when } t=0 \quad T=40$$

$$40 = -4 + Ae^0$$

$$A = 44$$

$$ii) \quad \frac{dT}{dt} = -kAe^{-kt}$$

$$= -k(T+4)$$

$$\therefore -0.3 = -k(26+4)$$

$$k = \frac{0.3}{30}$$

$$k = 0.01$$

$$iii) \quad T = -4 + 44e^{-0.01t}$$

$$6 = -4 + 44e^{-0.01t}$$

$$\frac{10}{44} = e^{-0.01t}$$

$$\ln \frac{10}{44} = -0.01t$$

$$t = \frac{-1}{0.01} \ln \frac{10}{44}$$

$$= 148.2 \text{ minutes}$$

Question 6

$$a) \int \frac{dx}{2x \sqrt{1 - (\ln x)^2}}$$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$= \frac{1}{2} \int \frac{du}{\sqrt{1 - u^2}}$$

$$= \frac{1}{2} \sin^{-1} u$$

$$= \frac{1}{2} \sin^{-1} (\ln x) + C$$

$$b) \quad 1) \quad \text{max height when } \dot{y} = 0$$

$$\dot{y} = 96 \sin \theta - gt$$

$$0 = 96 \sin \theta - gt$$

$$t = \frac{96 \sin \theta}{g}$$

$$\therefore y = 96 \left( \frac{96 \sin \theta}{g} \right) \sin \theta - \frac{g}{2} \left( \frac{96 \sin \theta}{g} \right)^2$$

Question 6

$$a) \int \frac{dx}{2x \sqrt{1 - (\ln x)^2}}$$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$= \frac{1}{2} \int \frac{du}{\sqrt{1 - u^2}}$$

$$= \frac{1}{2} \sin^{-1} u$$

$$= \frac{1}{2} \sin^{-1}(\ln x) + C$$

$$b) i) \text{ max height when } \dot{y} = 0$$

$$\dot{y} = 32 \sin \theta - gt$$

$$0 = 32 \sin \theta - gt$$

$$t = \frac{32 \sin \theta}{g}$$

$$\therefore \text{ max height} = 32 \left( \frac{32 \sin \theta}{g} \right) \sin \theta - \frac{1}{2} g \left( \frac{32 \sin \theta}{g} \right)^2$$

$$= \frac{1024 \sin^2 \theta}{g} - \frac{1024 \sin^2 \theta}{2g}$$

$$= \frac{512 \sin^2 \theta}{g}$$

$$ii) \text{ max distance when } \theta = 45^\circ \text{ and } y = 0$$

$$0 = 32 + \sin 45^\circ - \frac{1}{2} g t^2$$

$$0 = t \left( \frac{32}{\sqrt{2}} - \frac{1}{2} g t \right)$$

$$t = \cancel{0}, \frac{64}{\sqrt{2} g}$$

$$\therefore \text{max distance} = 32 \left( \frac{64}{\sqrt{2}g} \right) \cos 45^\circ$$

$$= \frac{1024}{g} \text{ metres}$$

$$\text{iii) } x = 32 \cos \theta \rightarrow t = \frac{x}{32 \cos \theta}$$

$$\therefore y = 32 \left( \frac{x}{32 \cos \theta} \right) \sin \theta - 5 \left( \frac{x}{32 \cos \theta} \right)^2$$

$$y = x \tan \theta - \frac{5x^2}{1024} (1 + \tan^2 \theta)$$

$$\text{when } x = 64, y = 30$$

$$30 = 64 \tan \theta - \frac{5 \times 64^2}{1024} (1 + \tan^2 \theta)$$

$$20 \tan^2 \theta - 64 \tan \theta + 50 = 0$$

$$\tan \theta = \frac{64 \pm \sqrt{64^2 - 4 \times 20 \times 50}}{40}$$

$$\theta = 62^\circ, 54^\circ$$

### Question 7

a) Step 1 : show true for  $n=1$

$$\text{LHS} = 5^1$$

$$= 5$$

$$\text{RHS} = 1 + 4$$

$$= 5$$

$$\text{LHS} \geq \text{RHS}$$

$\therefore$  true for  $n=1$

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Step 2: assume result true for  $n=k$

$$\text{i.e. } 5^k \geq 1 + 4k$$

Step 3: show result is true for  $n=k+1$

$$5^{k+1} = 5 \times 5^k$$

$$\geq 5(1 + 4k)$$

$$= 5 + 5 \cdot 4^k$$

$$\geq 1 + 4 \cdot 4^k$$

$$= 1 + 4^{k+1}$$

which is the required result

$\therefore$  true for  $n=k+1$  if true for  $n=k$

Step 4: as true for  $n=1$ , also true for  $n=2$ ,  $n=3$   
as true for  $n=2$ , also true for  $n=3$ ,  $n=4$   
and so on for all positive integers  $n$ .

b) let roots be  $\alpha, \beta, \alpha + \beta$

$$\therefore 2\alpha + 2\beta = -p$$

$$\alpha\beta + \alpha(\alpha + \beta) + \beta(\alpha + \beta) = q$$

$$\alpha\beta(\alpha + \beta) = -r$$

$$\text{or } \alpha + \beta = \frac{-p}{2} \quad (1)$$

$$\alpha^2 + \beta^2 + 3\alpha\beta = q \quad (2)$$

$$\alpha\beta(\alpha + \beta) = -r \quad (3)$$

sub (1) into (3)

$$\alpha_{\beta} \left( \frac{-p}{2} \right) = -r$$

$$\alpha\beta = \frac{2r}{\rho} \quad (4)$$

re arrange (2)

$$(\alpha + \beta)^2 + \alpha\beta = 9$$

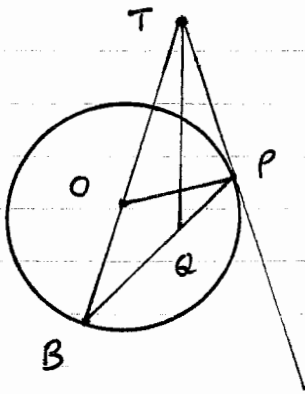
sub in (1) and (4)

$$\left(-\frac{p}{v}\right)^2 + \frac{2r}{p} = 9$$

$$\frac{p^2}{4} + \frac{2r}{p} = 9$$

$$10^3 + 8r = 499$$

2



$$\angle QTP = x$$

$$\angle BTQ = 2\alpha \quad (\text{given } QT \text{ bisects } \angle BTP)$$

$\angle TPO = 90^\circ$  (angle between radius and tangent)

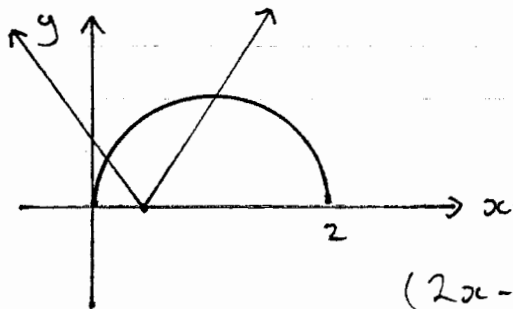
$$\angle TOP = 90^\circ - 2x \quad (\text{angle sum of triangle})$$

$\angle TBP = 45^\circ - x$  (angle at centre is double  
at circumference)

A  $\angle OPB = 45^\circ - x$  (equal angles in isosceles triangle)

$$\therefore \angle APB = 45^\circ + x \quad (\text{right angle})$$

d)



$$(2x-1)^2 = x(2-x)$$

$$5x^2 - 6x + 120$$

$$(5x-1)(x-1)=0$$

$$x = \frac{1}{5}, 1$$

$$\therefore 0 \leq x < \frac{1}{5}, \quad 1 < x \leq 2$$